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Inventor(s)

Shang; Yuanfang et al.

LIQUID TRANSFER DEVICE

Abstract

The application belongs to the field of liquid transfer, specifically to a liquid transfer device, which includes: a liquid path mechanism, including a conduit assembly and a driving member for driving liquid to flow in the conduit assembly; a housing assembly, including a bottom plate and a housing with an accommodating space, both the conduit assembly and driving member are installed on the housing; a support frame, including a rotating self-locking structure and a telescopic structure; the telescopic structure is installed on the bottom plate through the rotating self-locking structure; the rotating self-locking structure is used for driving the telescopic structure to rotate relative to the bottom plate within a preset angle range, and locking the telescopic structure to stop rotating once it reaches a preset position; and an electrical assembly, including a control assembly installed in the accommodating space, and the control assembly is connected with the driving member.

Inventors: Shang; Yuanfang (Shenzhen, CN), Guo; Xiaoliang (Shenzhen, CN), Huang; Minzhong (Shenzhen, CN), Ye; Peichen (Shenzhen, CN), Fang; Rulin (Shenzhen, CN), Liu; Fuming (Shenzhen, CN)

Applicant: SHENZHEN CELLBRI BIO-INNOVATION TECHNOLOGY CO., LTD.
(Shenzhen, CN)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a Continuation Application of PCT Application No. PCT/CN2023/133112 filed on Nov. 22, 2023, which claims the benefit of Chinese Patent Application No. 202211519527.6, filed on Nov. 30, 2022, titled “Liquid transfer device”, the contents of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The application belongs to the technical field of liquid transfer, in particular to a liquid transfer device.

BACKGROUND

[0003] Recently, the inventor(s) discovered that liquid transfer devices (such as fully enclosed liquid transfer instruments) are generally bulky and difficult to move, and must be positioned in a permanent position with large space for liquid transfer operation, which is inconvenient to use. Moreover, the existing liquid transfer device often lacks a built-in battery and requires an external power supply to function. The power cord limits its mobility and functionality, reducing the user experience.

SUMMARY OF THE INVENTION

[0004] The present application provides a retractable and foldable liquid transfer device that is small in size. And the portability of the liquid transfer device is improved.

[0005] In view of the above technical problems, an embodiment of the present application provides a liquid transfer device, including: [0006] a liquid path mechanism, comprising a conduit assembly and a driving member for driving liquid to flow in the conduit assembly; [0007] a housing assembly, comprising a bottom plate and a housing with an accommodating space, wherein both the conduit assembly and the driving member are installed on the housing; [0008] a support frame, comprising a rotating self-locking structure and a telescopic structure; the telescopic structure is installed on the bottom plate through the rotating self-locking structure; the rotating self-locking structure is used for driving the telescopic structure to rotate relative to the bottom plate within a preset angle range, and locking the telescopic structure to stop rotating once it reaches a preset position; and [0009] an electrical assembly, comprising a control assembly installed in the accommodating space, and the control assembly is connected with the driving member.

[0010] The liquid transfer device of the application can realize liquid flow to transfer liquid through the liquid path mechanism. Meanwhile, the telescopic structure of the application can be extended or shortened. The rotating self-locking structure can rotate and fold the telescopic structure, and lock it in place at a preset position (ensuring the stability of the liquid transfer operation of the device). Therefore, with the above-mentioned telescopic structure and rotating self-locking structure, the liquid transfer operation of the liquid transfer device can be ensured, while the volume of the liquid transfer device is also reduced. The liquid transfer device in the application is small in size and light in weight. When the telescopic structure is completely retracted and driven to be folded by the rotating self-locking structure, the length, width and height

of the liquid transfer device are 300 mm*220 mm*150 mm. It can be easily moved or placed into a biosafety cabinet or a clean workbench for use by just one person. Therefore, the liquid transfer device offered by the current application enhances its portability, guarantees regular liquid transfer operation, and also improves its structural stability and ease of mobility, making it easy for users to use the device in a small space.

[0011] The following figures and descriptions provide details for one or more embodiments of the application, and other features and advantages of the application will be apparent from the description, drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The application will be further illustrated with the drawings and embodiments.

[0013] FIG. 1 is a structural schematic diagram of the liquid transfer device according to an embodiment of the present application.

[0014] FIG. 2 is a structural schematic diagram of the electrical assembly of the liquid transfer device according to an embodiment of the present application.

[0015] FIG. 3 is a structural schematic diagram of the housing assembly of the liquid transfer device according to an embodiment of the present application.

[0016] FIG. 4 is a schematic diagram of the installation structure of the bubble sensor and driving member of the liquid transfer device according to an embodiment of the present application.

[0017] FIG. 5 is a structural schematic diagram of the liquid transfer device according to an embodiment of the present application when the telescopic structure is in an unfolded state. In this state, the rotating self-locking structure rotates to 90° and locks the telescopic structure.

[0018] FIG. 6 is a structural schematic diagram of the liquid transfer device according to an embodiment of the present application when the telescopic structure is in a retracted state. In this state, the rotating self-locking structure rotates to 0° and locks the telescopic structure.

[0019] FIG. 7 is a structural schematic diagram of the telescopic structure of the liquid transfer device according to an embodiment of the present application.

[0020] FIG. 8 is an explosive view of the telescopic structure of the liquid transfer device according to an embodiment of the present application.

[0021] FIG. 9 is a structural schematic diagram of the sliding block of the telescopic structure according to an embodiment of the present application.

[0022] FIG. 10 is a structural schematic diagram when the sliding plate of the telescopic structure is retracted in the sliding space and the clamping groove is clamped with the resisting buckle according to an embodiment of the application.

[0023] FIG. 11 is a structural schematic diagram when the sliding block of the telescopic structure slides upward with the sliding plate until the clamping groove is no longer clamped with the resisting buckle, according to an embodiment of the present application.

[0024] FIG. 12 is a structural schematic diagram when the sliding block of the telescopic structure is no longer clamped with the resisting buckle, and the sliding plate slides along the slide rail until the positioning assembly and the positioning hole are locked, according to an embodiment of the present application.

[0025] FIG. 13 is a structural schematic diagram when the sliding block of the telescopic structure slides upward with the sliding plate until it is out of the positioning hole, and the clamping groove has not been clamped with the resisting buckle, according to an embodiment of the present application.

[0026] FIG. 14 is a structural schematic diagram when the sliding block of the telescopic structure slides down with the sliding plate until the clamping groove is clamped with the resisting buckle,

according to an embodiment of the present application.

[0027] FIG. **15** is an explosive view of the rotating self-locking structure of the liquid transfer device according to an embodiment of the present application.

[0028] FIG. **16** is a structural schematic diagram in which the locking column of the rotating self-locking structure slides into the first positioning groove according to an embodiment of the present application.

[0029] FIG. **17** is a structural schematic diagram in which the locking column of the rotating self-locking structure slides out of the first positioning groove according to an embodiment of the present application.

[0030] FIG. **18** is a structural schematic diagram in which the locking column of the rotating self-locking structure slides into the first positioning groove and the arc buffer abuts against the stop block, according to an embodiment of the present application.

[0031] FIG. **19** is a structural schematic diagram in which the locking column of the rotating self-locking structure slides into the second positioning groove according to an embodiment of the present application.

[0032] FIG. **20** is a structural schematic diagram in which the locking column of the rotating self-locking structure slides into the second positioning groove and the arc buffer abuts against the stop block, according to an embodiment of the present application.

[0033] Reference signs in the drawings are as follows. [0034] **100**. Liquid path mechanism; **110**. Conduit assembly; **120**. Driving member; **121**. Pump head. **122**. First gasket; **123**. Pump shaft; **124**. Pump support; **125**. Motor; **130**. Bubble sensor; **131**. Second gasket; **140**. Liquid path mounting plate; **150**. Liquid path support rod; **200**. Housing assembly; **210**. Bottom plate; **220**. Housing; **221**. Accommodating space; **222**. Main device housing; **223**. Back plate; **224**. Bottom pad; **300**. Support frame; **310**. Telescopic structure; **10**. Support assembly; **101**. Support plate; **1011**. Sliding space; **1012**. Upper plate; **1013**. Lower plate; **102**. Slide rail; **103**. Positioning hole; **20**. Sliding assembly; **201**. Sliding plate; **202**. Positioning assembly; **2021**. Mounting hole; **2022**. First spring; **2023**. Sliding block; **20231**. Resisting surface; **20232**. Guide surface; **20233**. Clamping groove; **20234**. Guide groove; **2024**. Positioning pin; **2025**. Resisting buckle; **20251**. Through hole; **2026**. Guide rail; **203**. Sliding wheel; **204**. Front cover; **205**. Back cover; **30**. Top support block; **301**. Hook; **302**. Placement hole; **303**. Telescopic opening; **320**. Rotating self-locking structure; **1**. Rotating assembly; **11**. First rotating shaft; **111**. Positioning groove; **112**. Connecting segment; **113**. Cooperating segment; **114**. Accommodating groove; **115**. Arc baffle; **12**. Second rotating shaft; **2**. Bracket; **21**. First sliding hole; **22**. Second sliding hole; **23**. Sliding groove; **24**. Accommodating hole; **25**. Stop block; **26**. Bearing hole; **28**. First support plate; **29**. Second support plate; **3**. Lock assembly; **31**. Elastic self-locking member; **311**. Locking column; **312**. Lock buckle; **3121**. Sliding column; **313**. Second spring; **314**. Unlocking pin; **32**. Torque hinge; **4**. Bearing; **400**. Electrical assembly; **410**. Control assembly; **420**. Switch button; **430**. Battery; **440**. Power supply interface; **450**. Touch screen; **500**. Liquid bag.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

[0035] In order to make the technical problems, technical solutions and beneficial effects of the present application more clear, the application will be further explained in detail below with reference to the drawings and embodiments. It should be understood that the specific embodiments described here are only used to illustrate the application, rather than to limit the application.

[0036] It should be understood that the orientations or positional relationships indicated by the terms “upper”, “lower”, “left”, “right”, “front”, “rear” and “middle” and the like are based on the orientations or positional relationships shown in the attached drawings, only for convenience of describing the present invention and simplifying the description, and do not indicate or imply that the said device or element must have a specific orientation, be constructed or operated in a particular orientation, and therefore cannot be understood as a limitation of the present application.

[0037] As shown in FIGS. **1** to **6**, an embodiment of the present application provides a liquid

transfer device, including the following components.

[0038] A liquid path mechanism **100**, which includes a conduit assembly **110** and a driving member **120** for driving liquid to flow in the conduit assembly **110**. Optionally, as shown in FIG. **1**, the liquid path mechanism **100** is also installed on the bubble sensor **130** on the housing **220**, and the bubble sensor **130** is used to detect the liquid flow state in the conduit assembly **110**. Specifically, as shown in FIG. **4**, the driving member **120** includes, but is not limited to, a peristaltic pump. The pump support **2124** is fixed on the liquid path mounting plate **140** by screw connection, and the pump head **121** of the peristaltic pump is fixed on the pump support **2124** by screw connection. One end of the pump shaft **123** of the peristaltic pump is connected with the output shaft sleeve of the motor **125** and fixed on the output shaft of the motor **125** via a jackscrew. In this state, the pump shaft **123** can rotate together with the output shaft of the motor **125**, and the motor **125** is fixed on the liquid path mounting plate **140** via threaded connection. Moreover, the end of the pump shaft **123** away from the motor **125** should be clamped into the groove at the bottom of the pump head **121** of the peristaltic pump, that is, the pump shaft **123** can rotate together with the output shaft of the motor **125**, and then drive the peristaltic pump to rotate to squeeze the liquid flow in the conduit assembly **110**, so as to realize the liquid transfer. While the bubble sensor **130** is fixed on the liquid path mounting plate **140** via threaded connection, understandably, the bubble sensor **130** is used to monitor whether bubbles are mixed in the liquid in the conduit assembly **110** in real time, and then determine the liquid flow state. For example, the bubble sensor **130** detects that there are multiple bubbles in the liquid flowing in the conduit assembly **110** for one second, which indicates that no liquid has flowed in the conduit assembly **110**. This suggests that the liquid to be transferred has been exhausted (or will soon be exhausted). Furthermore, the liquid path mechanism **100** also includes a pressure sensor, which can detect the pressure of the liquid in the conduit assembly **110** to obtain a pressure detection result. In this application, the liquid transfer state can be determined according to the liquid flow state and pressure detection results, and then instructions such as stopping liquid transfer command and alarm can be issued according to the liquid transfer state.

[0039] A housing assembly **200**, which includes a bottom plate **210** and a housing **220** with an accommodating space **221**, and both the conduit assembly **110** and the driving member **120** are installed on the housing **220**. Understandably, as shown in FIG. **4**, the liquid path mounting plate **140** is fixed on the bottom plate **210** via the liquid path support rod **150**, allowing the installation structure of the liquid path mechanism **100** to be more stable. The housing **220** is provided with a first through hole for allowing the pump head **121** of the peristaltic pump to extend from the accommodating space **221** and clamp the conduit assembly **110**. The pump head **121** and the housing **220** are connected sealingly via the first gasket **122** to prevent water, oil or other impurities from entering the accommodating space **221** from the installation position between them. Furthermore, the housing **220** is provided with a second through hole for enabling the bubble sensor **130** to extend out of the accommodating space **221** and clamp the conduit assembly **110** for alignment, so as to measure the liquid flow state. And the bubble sensor **130** and the housing **220** are connected sealingly via the second gasket **131** to prevent water, oil and other impurities from entering the accommodating space **221** from the installation position between them. Optionally, as shown in FIG. **3**, the housing **220** includes a main device housing **222** installed on the bottom plate **210** and having an opening, and a back plate **223** installed at the opening position of the main device housing **222** and connected to the back plate **223** of the bottom plate **210**. The main device housing **222**, back plate **223** and bottom plate **210** enclose to form the accommodating space **221**, and the control assembly **410** is installed on the bottom plate **210** and located in the accommodating space **221**. In the embodiment shown in FIG. **3**, the housing **220** further includes a bottom pad **224**, and the bottom pad **224** is fixed on the bottom of the bottom plate **210** via threaded connection, and the main device housing **222** is fixed on the bottom plate **210** via threaded connection. The back plate **223** is fixed on main device housing **222** and bottom plate **210** via

threaded connection, and the housing assembly **200** is used to support and protect the whole liquid transfer device.

[0040] A support frame **300**, which includes a rotating self-locking structure **320** and a telescopic structure **310**. As shown in FIGS. **5** and **6**, the telescopic structure **310** is installed on the bottom plate **210** through the rotating self-locking structure **320**. The rotating self-locking structure **320** is used to drive the telescopic structure **310** to rotate within a preset angle range relative to the bottom plate **210**, and lock the telescopic structure **310** to stop rotating when it rotates to a preset position. Specifically, the support frame **300** is mainly used for placing a liquid bag **500** or a bottle (such as a bottle made of plastic, glass, etc.) that needs liquid transfer, and the shapes of the liquid bag **500** and the bottle are not restricted. The telescopic and foldable rotating self-locking structure **320**, which is composed of the telescopic structure **310** and the rotating self-locking structure **320**, is designed to reduce the device's volume while also enabling the liquid transfer device to operate conveniently.

[0041] An electrical assembly **400**, which includes a control assembly **410** installed in the accommodating space **221**, and the control assembly **410** is connected with the driving member **120**. As shown in FIGS. **1** and **2**, the electrical assembly **400** is fixedly installed on the housing assembly **200**, and the control assembly **410** in the electrical assembly **400** is used to control the operation of device. For example, the control assembly **410** is connected with the driving member **120**, the bubble sensor **130**, the pressure sensor, etc., and is used to control the operation of various components. Optionally, the electrical assembly **400** further includes a touch screen **450** installed on the housing **220** and connected to the control assembly **410**. The touch screen **450** is a man-machine interactive interface, which can be used for parameter adjustment and instruction operation, etc. The touch screen **450** is connected with the control assembly **410**, so that the liquid transfer parameters and liquid transfer modes can be set via the touch screen **450**, and then different liquid transfer operations can be performed through the controller according to the above settings.

[0042] Furthermore, as shown in FIGS. **1** and **2**, the electrical assembly **400** further includes a switch button **420** installed on the housing **220** and connected to the control assembly **410**. In this way, with the trigger of the switch button **420**, the controller can start or turn off the liquid transfer device. Optionally, the electrical assembly **400** further includes a battery **430** connected to the control assembly **410** and a power supply interface **440** connected to the battery **430** and the control assembly **410**. Specifically, the battery **430** can supply power to the liquid transfer device when the power supply is not turned on. The liquid transfer device of this application has a built-in battery **430**, which can enable the liquid transfer device to work for **48** hours in the standby state and continuously transfer liquid for **4** hours in the liquid transfer state. In this application, the power supply interface **440** can be connected with an external power supply to supply power to the liquid transfer device, and can also charge the battery **430**. Understandably, the electrical assembly **400** of the present application further includes a prompting device connected to the control assembly **410**, which may be a buzzer that emits a prompting sound, prompts when the liquid transfer is complete, or is used when an abnormal alarm is triggered. Understandably, the prompting device may also be a light prompting device capable of emitting lights with different colors and flashing frequencies, as well as performing various sorts of light prompting under varying prompting conditions.

[0043] The liquid transfer device of the present application can realize liquid flow to transfer liquid through the liquid path mechanism **100**. Meanwhile, the telescopic structure **310** of the present application can be lengthened or shortened, and the rotating self-locking structure **320** can rotate and fold the telescopic structure **310**, and lock it at the preset position (ensuring the stability of the device for liquid transfer operation). Furthermore, with the above-mentioned telescopic structure **310** and rotating self-locking structure **320**, the volume of the liquid transfer device can be reduced while assuring that it can conduct liquid transfer operations. The liquid transfer device in this application is small in size and light in weight. When the telescopic structure **310** is completely retracted and is folded by the rotating self-locking structure **320**, the size, length, width and height

of the liquid transfer device are 300 mm*220 mm*150 mm, which can be easily moved by a single person or put into a biosafety cabinet or a clean workbench for operation. Therefore, the liquid transfer device provided by the present application improves the portability of the liquid transfer device, ensures the normal liquid transfer operation, and also improves the moving convenience and structural stability, which is convenient for users to use the liquid transfer device in a small space. The liquid transfer device in this application can automatically transfer the liquid in the bottle or the liquid bag **500** to the targeted container through the liquid path mechanism **100**. The flow speed range of the liquid transfer device in this application is 0-200 ml/min, and the volume of the transferred liquid can be set as required.

[0044] The working principle of the liquid transfer device in an embodiment of the application is as follows. When the liquid transfer device is powered on or the battery **430** has enough power, press the switch button **420**, wait for a few seconds, and the liquid transfer device will turn on and finish the self-test, allowing regular functioning to resume. In this state, as shown in FIG. **1**, a large liquid bag **500** with liquid to be transferred is hung on the support frame **300** (or a bottle with liquid to be transferred is placed in the placement hole **302** of the top support block **30** at the top of the support frame **300**). Understandably, as shown in FIG. **5**, the top of the placement hole **302** is provided with an inclined plane, so that the opening of the inverted bottle can fit and fit with the transition inclined plane of the bottle body, and the installation is stable. And the placement hole **302** has a telescopic opening **303**, and the aperture size of the placement hole **302** can be adjusted within a certain range through the setting of the telescopic opening, so as to be suitable for bottles of different sizes). Then the conduit assembly **110** is clamped into the bubble sensor **130** and the pump head **121** of the peristaltic pump, and the target container (such as a small liquid bag) connected to the conduit assembly **110** is placed in a preset position next to the liquid transfer device. Then, select the volume and speed of liquid transfer on the operation interface of the touch screen **450**. And then click "Start" on the touch screen **450**, the liquid transfer device will start to transfer the liquid in the large liquid bag **500** into the small liquid bag. When the buzzer on the liquid transfer device makes the "didi" sound, it indicates that the liquid transfer is complete.

[0045] In an embodiment, as shown in FIGS. **7** to **9**, the telescopic structure **310** includes the following components.

[0046] A support assembly **10**, which includes a support plate **101** with a sliding space **1011**, a slide rail **102** arranged on the inner side wall of the sliding space **1011**, and a positioning hole **103** arranged on the slide rail **102** and communicating with the sliding space **1011**. Optionally, the support plate **101** includes an upper plate **1012** and a lower plate **1013** connected to the upper plate **1012**, and the sliding space **1011** is formed between the upper plate **1012** and the lower plate **1013**. The upper plate **1012** and the lower plate **1013** can be fixedly connected by snap connection, screw connection, welding and the like, or the upper plate **1012** and the lower plate **1013** can be integrally formed.

[0047] A sliding assembly **20**, which includes a sliding plate **201** arranged in the sliding space **1011** and sliding telescopically along the slide rail **102**, and a positioning assembly **202** arranged on the sliding plate **201**. The positioning assembly **202** is used for locking with the positioning hole **103** when the sliding plate **201** slides towards the first end of the slide rail **102** to stop the sliding of the sliding plate **201**. The positioning assembly **202** is also used to unlock the positioning hole **103** when the sliding plate **201** slides towards the second end of the slide rail **102**.

[0048] As shown in FIG. **7**, the support assembly **10** includes two slide rails **102** arranged on opposite sides of the sliding plate **201**. The sliding plate **201** slides along the slide rail **102** arranged on both sides of the sliding plate **201** to realize the telescopic movement between the sliding plate **201** and the support plate **101**. The sliding assembly **20** includes two positioning assemblies **202** arranged on opposite sides of the sliding plate **201**. The positioning holes **103** are symmetrically arranged on two slide rails **102**. The positioning assemblies **202** are symmetrically arranged on the opposite sides of the sliding plate **201**, so as to cooperate with the positioning hole **103** to lock or

unlock the sliding plate **201**. Furthermore, the sliding assembly **20** further includes a front cover **204** and a back cover **205** respectively installed on opposite sides of the sliding plate **201**. The front cover **204**, the sliding plate **201** and the back cover **205** are fixedly connected. Specifically, the front cover **204** and back cover **205** can be fixedly connected by means of snap connection, screw connection, welding and the like. Or the front cover **204** and the back cover **205** can be integrally formed. However, the sliding plate **201** needs to be fixedly installed between the front cover **204** and the back cover **205** (for example, the sliding plate **201** can be fixed on the back cover **205** by screws). The sliding plate **201**, front cover **204** or back cover **205** can be arranged with a top support block **30**. Components required for operations such as liquid transfer can be installed on the top support block **30** (such as the hook **301** required when the liquid bag **500** for liquid transfer is hung, such as the hook **301** of the retractable and foldable support frame **300** shown in FIG. 7, with a limit load of 5 kg). And as shown in FIG. 5, the top support block **30** is provided with a placement hole **302** with a telescopic opening **303**, and the top support block **30** can be designed with the same height as the first support plate **28** and the second support plate **29**, so as to enable the top support block **30**, first support plate **28** and second support plate **29** to jointly support the locked telescopic structure **310** when the telescopic structure **310** rotates to 0 degrees as shown in FIG. 6. Furthermore, after the support plate **101** is installed on the bottom plate **210**, the components installed on the top support block **30** can be driven to move synchronously by the movement of the sliding plate **201** relative to the support plate **101**, and then the distance between the bottom plate **210** and the components can be adjusted.

[0049] The telescopic structure **310** of this embodiment allows for a length expansion adjustment between the sliding plate **201** and support plate **101**. Specifically, the sliding plate **201** of the telescopic structure **310** can be shrunk in the sliding space **1011** of the support plate **101**, thereby reducing the size of the whole liquid transfer device. However, when the length between the movable plate and the support plate **101** needs to be adjusted according to the needs of the operation related to liquid transfer, the sliding plate **201** retracted in the sliding space **1011** slides along the slide rail **102**. In this state, the length between the sliding plate **201** and the support plate **101** is extended. And when the sliding plate **201** slides to the positioning assembly **202** and locks with the positioning hole **103**, the sliding plate **201** can be fixed at this position, thus ensuring the structural stability of the liquid transfer device when performing operations such as liquid transfer. In this application, the telescopic structure **310** enables the combined length of the sliding plate **201** and supporting plate **101** to be extended to a preset length (after retracted, the length of the sliding plate **201** and supporting plate **101** can be reduced by about half, and the preset length may be 100 mm), and the locking can be realized by the positioning hole **103** and the positioning assembly **202** at the preset length. The telescopic structure **310** of the present application improves the portability and structural stability of the liquid transfer device, and improves its moving convenience while ensuring regular liquid transfer operations, which is also convenient for users to use it for liquid transfer operation in a small space.

[0050] In an embodiment, as shown in FIGS. 7 to 8, the sliding assembly **20** further includes a sliding wheel **203** arranged on the sliding plate **201**, and the sliding plate **201** slides along the slide rail **102** in a telescopic manner through the sliding wheel **203**. That is, the two sliding wheels **203** are respectively fixed on the opposite sides of the sliding plate **201** by screws, and the two sliding wheels **203** can slide along the two slide rail **102** at the same time, thereby driving the sliding plate **201** to move smoothly in the sliding space **1011**.

[0051] In an embodiment, the positioning assembly **202** includes a resisting buckle **2025** arranged on the sliding plate **201**, a mounting hole **2021** arranged on the sliding plate **201** and communicating with the sliding space **1011**, and a first spring **2022** and sliding block **2023** both installed in the mounting hole **2021**. One end of the sliding block **2023** abuts against the first spring **2022**, and the other end of the sliding block **2023** extends into the sliding space **1011** from the mounting hole **2021**. One side of the sliding block **2023** near the first end of the slide rail **102** is

provided with a resisting surface **20231**; one side of the sliding block **2023** near the second end of the slide rail **102** is provided with a guide surface **20232**. The first spring **2022** is in a compressed state. In this way, as shown in FIG. **12**, when the sliding block **2023** slides to be opposite to the positioning hole **103**, under the elastic force of the first spring **2022**, the sliding block **2023** will extend into the positioning hole **103** to catch the sliding plate **201** and lock it. In this state, the sliding block **2023** is provided with a resisting surface **20231** on the side near the first end of the slide rail **102** (i.e., the lower end of the slide rail **102** shown in FIG. **10**). In this way, as shown in FIG. **12**, when the sliding plate **201** is pushed toward the first end of the slide rail **102**, because the resisting surface **20231** of the sliding block **2023** abuts against the inner wall of the positioning hole **103**, the sliding plate **201** cannot move. Since the side of the sliding block **2023** near the second end of the slide rail **102** (i.e., the upper end of the slide rail **102** shown in FIG. **10**) is provided with a guide surface **20232** (alternatively, the guide surface **20232** is an inclined surface or an arc surface, as long as the sliding block **2023** can slide out along the guide surface **20232** when the sliding plate **201** is pushed toward the second end of the slide rail **102**). In this state, if the sliding plate **201** is pushed toward the second end of the slide rail **102**, under the guidance of the guide surface **20232**, the sliding block **2023** can gradually slide out of the positioning hole **103** along the guide surface **20232** as shown in FIG. **13**, thus unlocking the sliding plate **201**.

Furthermore, a guide rail **2026** perpendicular to the slide rail **102** is arranged on the inner side wall of the mounting hole **2021**, and a guide groove **20234** slidably connected with the guide rail **2026** is arranged on the sliding block **2023**. That is, the sliding block **2023** moves along the guide rail **2026** via the guide groove **20234**, which can ensure the accuracy of the moving direction. In an embodiment, as shown in FIGS. **8** and **10**, the positioning assembly **202** further includes a positioning pin **2024** connected to the sliding block **2023**, and the first spring **2022** is sleeved at one end of the positioning pin **2024** away from the sliding block **2023**. That is, the positioning pin **2024** is fixed on the sliding block **2023** via threaded connection, and the positioning pin **2024** passes through the first spring **2022** and extends into a pin hole (not shown) provided on the sliding plate **201**. The arrangement of the positioning pin **2024** can ensure that the first spring **2022** does not deform during the extension and retraction, thus ensuring the stability of its elasticity.

[0052] The resisting buckle **2025** is provided with a through hole **2025**, and the sliding block **2023** extends into the sliding space **1011** from the mounting hole **2021** through the through hole **2025**. A clamping groove **20233** adapted to the resisting buckle **2025** is formed on the resisting surface **20231** of the sliding block **2023**. As shown in FIGS. **8** and **10**, a resisting buckle **2025** arranged on the sliding plate **201** is slidably connected with the slide rail **102**. After the sliding block **2023** extends into the sliding space **1011** from the mounting hole **2021** through the through hole **2025**, the clamping groove **20233** can be clamped with the resisting buckle **2025**, but they may not be clamped. As shown in FIG. **10**, the first spring **2022** is in a compressed state when the clamping groove **20233** is clamped with the resisting buckle **2025**. Understandably, when the clamping groove **20233** is clamped with the resisting buckle **2025**, the sliding block **2023** is in contact with the inner side wall of the slide rail **102**. When the sliding plate **201** is completely retracted in the sliding space **1011**, the clamping groove **20233** and the resisting buckle **2025** are in a clamping state.

[0053] Furthermore, in the direction parallel to the slide rail **102**, the width of the through hole **2025** is greater than the maximum width of the relative moving segment of the sliding block **2023**. The relative moving segment includes a first length segment opposite to the clamping groove **20233**, and a second length segment connected with the first length segment and wider than the first length segment. That is, the relative moving segment refers to the length segment that the sliding block **2023** needs to pass through the through hole **2025** when it moves in the mounting hole **2021** (including the length that the sliding block **2023** needs to move in the through hole **2025** when it extends into or slides out of the positioning hole **103**). The width of the through hole **2025** needs to allow the sliding block **2023** to move in the through hole **2025**, so that it can switch

between the two states of clamping and not clamping with the resisting buckle **2025**. Specifically, as shown in FIG. **11**, when the sliding plate **201** needs to be stretched, the sliding plate **201** is pulled upward. In this state, the sliding block **2023** moves upward in the through hole **2025**, and the clamping groove **20233** and the resisting buckle **2025** switch from the clamping state to the non-clamping state. In this state, if the sliding plate **201** is continuously pulled upward, then when the sliding plate **201** slides along the slide rail **102** to a position where the positioning assembly **202** is opposite to the positioning hole **103**, the first spring **2022**, which is still in compression, will continue to push the sliding block **2023** into the positioning hole **103** by its elastic force, and then lock the sliding plate **201** in the positioning hole **103** as shown in FIG. **12**.

[0054] Understandably, as shown in FIG. **12**, the sliding plate **201** is locked. And as shown in FIG. **13**, after the sliding plate **201** is continuously pulled upward to lead the sliding plate **201** out of the positioning hole **103** via the guide surface **20232**, so as to unlock the sliding plate **201**, in this state, since the sliding block **2023** is in contact with the inner wall of the slide rail **102**, and the clamping groove **20233** is opposite to the resisting buckle **2025**, if the sliding plate **201** is pushed down. Then as shown in FIG. **14**, the clamping groove **20233** and the resisting buckle **2025** switch from the non-clamping state to the clamping state, so when the sliding plate **201** is pushed downwards, even if it moves to the position where the sliding block **2023** is opposite to the positioning hole **103**, because the clamping groove **20233** is locked on the resisting buckle **2025** and cannot move towards the positioning hole **103**, the sliding plate **201** can continue to move downwards without being locked by the positioning hole **103**, and finally the sliding plate **201** can be completely retracted in the sliding space **1011** (as shown in FIG. **10**).

[0055] In a specific embodiment, the locking process of the telescopic structure **310** is as follows.

[0056] When the sliding plate **201** of the telescopic structure **310** is completely retracted in the sliding space **1011**, that is, the telescopic structure **310** is in an unextended state, as shown in FIG. **10**. In this state, the clamping groove **20233** is clamped with the resisting buckle **2025**. When the telescopic structure **310** needs to be extended, the sliding plate **201** is pulled upward. In this state, the sliding plate **201** moves upward (the resisting bucket **2025** does not move in this state). When the sliding plate **201** moves upward by a certain distance (for example, about 1.5 mm), the sliding block **2023** will touch the resisting bucket **2025**. In this state, the clamping groove **20233** and resisting buckle **2025** are no longer clamped, and then the sliding block **2023** drives the resisting buckle **2025** to move upward together. In this state, the sliding block **2023** tends to move outward under the elastic force of first spring **2022** in the compressed state (as shown in FIG. **11**). When the sliding plate **201** continues to be stretched upward to a preset length (for example, 100 mm), the sliding block **2023** is opposite to the positioning hole **103**. In this state, the sliding block **2023** will quickly enter the positioning hole **103** under the elastic force of the first spring **2022**, with a sound of “click”. This indicates that the sliding plate **201** has been locked by the positioning hole **103**. In this state, the telescopic structure **310** is in the extended and self-locking state, and the downward sliding will be hindered (as shown in FIG. **12**).

[0057] The unlocking process of the telescopic structure **310** is as follows.

[0058] When the telescopic structure **310** is in the extended state (as shown in FIG. **12**) and in the self-locking state, it cannot move downward. In this state, when the sliding plate **201** is pulled upward, the guide surface **20232** of the sliding block **2023** will convert the pulling force into a thrust force that pushes the sliding block **2023** to further compress the first spring **2022** (the thrust force direction is opposite to the elastic force direction of the first spring **2022**). In this way, under the guidance of the guide surface **20232** which generates thrust, the sliding block **2023** can gradually slide out of the positioning hole **103** along the guide surface **20232** as shown in FIG. **13**, thus unlocking the sliding plate **201**. In this state, when the sliding plate **201** is pushed back for a short distance, the resisting buckle **2025** will be clamped into the clamping groove **20233** under the sliding block **2023** (as shown in FIG. **14**), and the sliding block **2023** will not slide outward or inward due to the blocking of the resisting buckle **2025**. In this state, the sliding plate **201** can be

pushed back to the position shown in FIG. 10 smoothly.

[0059] In an embodiment, as shown in FIGS. 15 to 20, the rotating self-locking structure 320 includes the following components.

[0060] A rotating assembly 1, which includes a first rotating shaft 11 and a second rotating shaft 12. The first rotating shaft 11 and second rotating shaft 12 are fixedly installed on opposite sides of the support plate 101 and coaxially arranged. Specifically, both the first rotating shaft 11 and second rotating shaft 12 are fixedly connected with the support plate 101 of the telescopic structure, for example, by means of screw connection.

[0061] A bracket 2; the second rotating shaft 12 is rotatably connected to the bracket 2. In an embodiment, the bracket 2 is provided with a bearing hole 26, and the rotating assembly 1 further comprises a bearing 4, and the second rotating shaft 12 is rotatably installed in the bearing hole 26 via the bearing 4. That is, the bearing 4 is tightly fixed in the bearing hole 26, and the second rotating shaft 12 is tightly fixed in the inner ring of the bearing 4 and can rotate freely.

Furthermore, as shown in FIG. 1, the bracket 2 includes a first support plate 28 and a second support plate 29, both installed on the bottom plate 210. The second rotating shaft 12 is rotatably mounted on the second support plate 29, and the first rotating shaft 11 is rotatably mounted on the first support plate 28 via the torque hinge 32. That is, one side of the support plate 101 is rotatably connected to the second support plate 29 via the second rotating shaft 12, the other side of the support plate 101 is rotatably connected to the first support plate 28 via the first rotating shaft 11 and torque hinge 32, thus the support plate 101 can rotate relative to the bracket 2 together with the first rotating shaft 11 and second rotating shaft 12.

[0062] A lock assembly 3, which includes an elastic self-locking member 31 slidably connected with the bracket 2 and a torque hinge 32 installed on the bracket 2. The first rotating shaft 11 is rotatably connected with the bracket 2 via the torque hinge 32. The first rotating shaft 11 is provided with a positioning groove 111, and the elastic self-locking member 31 is provided with a locking column 311 adapted to the positioning groove 111. The support plate 101 rotates around the first rotating shaft 11 relative to the bottom plate 210 within a preset angle range. The locking column 311 is used to slide into the positioning groove 111 when the support plate 101 rotates around the first rotating shaft 11 to a preset position, thereby locking the support plate 101 to limit the rotation of the support plate 101 around the first rotating shaft 11, and the locking column 311 is also used to unlock the support plate 101 when sliding out of the positioning groove 111. As shown in FIG. 16, the locking column 311 is positioned in the positioning groove 111. In this state, the first rotating shaft 11 is blocked by the locking column 311 and cannot rotate. Therefore, the support plate 101 does not need to rotate with the first rotating shaft 11, so the support plate 101 is locked. As shown in FIG. 17, the locking column 311 slides out along the positioning groove 111. When the locking column 311 completely slides out of the positioning groove 111, the first rotating shaft 11 is no longer clamped by the locking column 311. In this state, the first rotating shaft 11 can drive the support plate 101 to rotate relative to the bracket 2.

[0063] In this embodiment, the rotating self-locking structure 320 can realize the self-locking and folding of the support plate 101 (I.e., the telescopic structure). That is, the support plate 101 can rotate and self-lock between the working position and non-working position via the above-mentioned rotating self-locking structure 320, so that the telescopic structure can be locked at the working position, and then the normal operations such as liquid transfer can be performed, thus ensuring the structural stability of the liquid transfer device during the operations such as liquid transfer. When there is no need for operations such as liquid transfer, the telescopic structure is locked and folded in the non-working position by rotating, thus reducing the volume of the whole liquid transfer device. The rotating self-locking structure 320 of the present application can realize the folding and self-locking of the telescopic structure within a preset angle range (such as an angle range of 0°-90°). The rotating self-locking structure 320 of the present application improves the portability and structural stability of the liquid transfer device, improves its moving convenience

while ensuring the normal liquid transfer operation, and is also convenient for users to use it for liquid transfer operation in a small space.

[0064] In an embodiment, as shown in FIGS. 15 to 20, the elastic self-locking member 31 includes a lock buckle 312, a second spring 313 and an unlocking pin 314. The locking column 311 is arranged on the lock buckle 312; the bracket 2 is provided with a first sliding hole 21 and a second sliding hole 22 which are parallel to each other. The lock bucket 312 is provided with a sliding column 3121 that passes through the first sliding hole 21 from the first side of the bracket 2 and slides along it, and the unlocking pin 314 passes through the second sliding hole 22 from the second side of the bracket 2 away from the first side and is connected with the lock bucket 312. The second spring 313 is sleeved on the unlocking pin 314 and connected between the bracket 2 and the lock buckle 312 via compression. Specifically, the number of the first sliding holes 21 can be determined as required. For example, the number of the sliding columns 3121 on the lock buckle 312 shown in FIG. 15 is two. In this state, the number of the first sliding holes 21 needs to be set to two accordingly, and all the first sliding holes 21 are parallel to each other. That is, the sliding column 3121 of the lock buckle 312 passes through the first sliding hole 21 from the direct first side, and the unlocking pin 314 passes through the second sliding hole 22 on the bracket 2 from the second side. And the end of the unlocking pin 314 passing through the second sliding hole 22 is fixed on the lock buckle 312 after the second spring 313 is sleeved, so that the lock buckle 312 and the unlocking pin 314 are fixed into a whole. Since the first sliding hole 21 and second sliding hole 22 are parallel, when the unlocking pin 314 slides along the second sliding hole 22, the sliding column 3121 will also drive the lock buckle 312 to slide along the first sliding hole 21.

Furthermore, the locking column 311 arranged on the lock buckle 312 will also move in the direction parallel to the first sliding column 3121. Therefore, in this application, the positioning groove 111 (the locking column 311 slides along the positioning groove 111) and the first sliding hole 21 also need to be arranged in parallel. As shown in FIG. 16, the locking column 311 is positioned in the positioning groove 111. In this state, the first rotating shaft 11 is clamped by the locking column 311 and cannot rotate. Therefore, the support plate 101 does not need to rotate with the first rotating shaft 11, so the support plate 101 is locked. When unlocking the support plate 101, as shown in FIG. 17, the unlocking pin 314 is pulled away from the bracket 2. In this state, the unlocking pin 314 slides along the second sliding hole 22, thereby driving the sliding column 3121 and the lock buckle 312 to slide along the first sliding hole 21, which further drives the locking column 311 on the lock buckle 312 to slide out along the positioning groove 111. When the locking column 311 completely slides out of the positioning groove 111, the first rotating shaft 11 is no longer clamped by the locking column 311. In this state, the first rotating shaft 11 can drive the support plate 101 to rotate relative to the bracket 2.

[0065] In an embodiment, as shown in FIG. 17, the bracket 2 is provided with a sliding groove 23 adapted to the lock buckle 312. The lock buckle 312 is slidably installed in the sliding groove 23. That is, the shape of the sliding groove 23 is adapted to the outer contour of the lock bucket 312, so that the lock bucket 312 can maintain its stability during sliding. The lock buckle 312 in FIG. 17 is similar to a square (there is a protruding locking column 311 on the square), so the sliding groove 23 is also similar to a square. During the sliding of the sliding column 3121 along the first sliding hole 21, the lock buckle 312 itself will also slide along the sliding groove 23, thus the sliding direction of the lock buckle 312 along the sliding groove 23 is parallel to the first sliding hole 21.

[0066] In an embodiment, the lock buckle 312 is provided with a groove (not shown) adapted to the second spring 313, and the bottom of the groove is provided with a connecting hole (not shown). The second spring 313 is placed in the groove, and the unlocking pin 314 passes through the second spring 313 and is connected with the connecting hole. That is, in this embodiment, the second spring 313 is placed in a groove on the lock buckle 312, and the groove is preferably a circular groove adapted to the second spring 313. The unlocking pin 314 passes through the second sliding hole 22 (clearance fit with the second sliding hole 22) and the second spring 313, and is

fixed in the connecting hole via threaded connection.

[0067] In an embodiment, as shown in FIGS. **16** to **20**, the first rotating shaft **11** includes a connecting segment **112** connected to the support plate **101** and a cooperating segment **113** connected to the connecting segment **112**. The cooperating segment **113** is provided with an accommodating groove **114**, the bracket **2** is provided with an accommodating hole **24**, one end of the torque hinge **32** is interference fitted in the accommodating hole **24**, and the other end of the torque hinge **32** is interference fitted in the accommodating groove **114**. Specifically, one end of the torque hinge **32** is fixed in the accommodating groove **114** of the first rotating shaft **11** by interference fit, and the other end of the torque hinge **32** is fixed in the accommodating hole **24** of the bracket **2** by interference fit. Moreover, the cooperating segment **113** of the first rotating shaft **11** is inserted into the accommodating hole **24** of the bracket **2** (first support plate **28**) and fits with the accommodating hole **24** in clearance, so that the first rotating shaft **11** can rotate in the accommodating hole **24**.

[0068] In an embodiment, as shown in FIGS. **18** and **20**, the cooperating segment **113** is provided with an arc baffle **115**, which is inserted into the accommodating hole **24** and slidably connected between the torque hinge **32** and the inner side wall of the accommodating hole **24**. A stop block **25** is arranged on the inner side wall of the accommodating hole **24**. The stop block **25** is used to restrict the first rotating shaft **11** from driving the support plate **101** to rotate when it abuts against the arc baffle **115**. That is, the arc buffer **115** is inserted into the accommodating hole **24**; and both the arc baffle **115** and torque hinge **32** can rotate in the accommodating hole **24**, at least two stop blocks **25** can be arranged on the inner side wall of the accommodating hole **24**. As shown in FIG. **18**, the first rotating shaft **11** rotates clockwise until the arc baffle **115** abuts against the stop block **25** above. In this state, the first rotating shaft **11** cannot continue to rotate clockwise due to the limitation of the stop block **25**. Similarly, as shown in FIG. **20**, the first rotating shaft **11** rotates counterclockwise until the arc baffle **115** abuts against the stop block **25** below. In this state, the first rotating shaft **11** cannot continue to rotate counterclockwise due to the limitation of the stop block **25**. The upper and lower stop block **25** in FIGS. **18** and **20** are designed to be 180 degrees apart in the circumferential direction of the accommodating hole **24**, and the length of the arc baffle **115** is designed to be equal to a quarter of the circumference of the accommodating hole **24**. Therefore, the actual rotation angle range of the arc baffle **115** in the accommodating hole **24** is only 0 to 90 degrees. Therefore, in the embodiments shown in FIGS. **18** and **20**, the first rotating shaft **11** can only drive the support plate **101** to rotate 90 degrees. Understandably, in this application, if it is required to modify the rotation angle range of the support plate **101** in order to change the rotatable angle of the support plate **101**, the length of the arc baffle **115** and the position arrangement of the stop block **25** can be adjusted, which is not repeated here.

[0069] In an embodiment, as shown in FIGS. **16** to **20**, at least two positioning grooves **111** are arranged on the cooperating segment **113**. At least one of the positioning groove **111** is aligned with the locking column **311** when the stop block **25** abuts against the arc baffle **115**. Specifically, the cooperating segment **113** is provided with two or more positioning grooves **111** to realize multi-level positioning of the support plate **101**. In the embodiments shown in FIGS. **16** to **20**, only two positioning grooves **111** are provided, and as shown in FIG. **18**, when the first rotating shaft **11** rotates clockwise until the arc baffle **115** abuts against the stop block **25** above, the first rotating shaft **11** cannot continue to rotate clockwise due to the limitation of the stop block **25**. In this state, the locking column **311** on the lock buckle **312** will be opposite to one of the positioning groove **111**. If the unlocking pin **314** is not controlled, as shown in FIG. **16**, it will slide into the positioning groove **111** by itself under the pulling force of the second spring **313**, and fix the support plate **101** at the rotation angle. Similarly, as shown in FIG. **20**, when the first rotating shaft **11** rotates counterclockwise until the arc baffle **115** abuts against the stop block **25** below, the first rotating shaft **11** cannot continue to rotate counterclockwise due to the limitation of the stop block **25**. In this state, the locking column **311** on the lock buckle **312** will be opposite to another positioning

groove **111**. If the unlocking pin **314** is not controlled, as shown in FIG. **19**, will slide into another positioning groove **111** by itself under the pulling force of the second spring **313**, so as to position the support plate **101** at another angle.

[0070] In an embodiment, the self-locking and unlocking process of the rotating self-locking structure **320** are as follows.

[0071] As shown in FIG. **16**, the support plate **101** of the rotating self-locking structure **320** is in a 0° self-locking state, and the lock buckle **312** is clamped into the first positioning groove **111** of the first rotating shaft **11**. In this state, the first rotating shaft **11** cannot rotate, and when the unlocking pin **314** fixed with the lock buckle **312** is pulled out, the locking column **311** will slide out along the positioning groove **111** until the positioning groove **111** slides by a preset distance (for example, 3.25 mm), the locking column **311** of the lock buckle **312** is separated from the positioning groove **111** of the first rotating shaft **11** (as shown in FIG. **17**). In this state, the first rotating shaft **11** can be rotated counterclockwise (clockwise, the door baffle has been blocked by the stop block **25** above and cannot be rotated, as shown in FIG. **18**), until it is rotated from 0° to 90° , since the arc baffle **115** cannot be rotated counterclockwise because it is blocked by the stop block **25** below (as shown in FIG. **20**), the unlocking pin **314** is released in this state, and under the force of the second spring **313**, the locking column **311** of the lock buckle **312** will be clamped into another positioning groove **111** of the first rotating shaft **11** (as shown in FIG. **19**). In this state, the second rotating shaft **12** will be locked in this position again. The process of unlocking, rotating and re-locking in the position shown in FIG. **19** is the same, and will not be repeated here.

[0072] The above are merely the preferred embodiments of the liquid transfer device of this application, and are not intended to limit the application. Any modification, equivalent substitution and improvement made within the spirit and principle of this application shall be included in the protection scope of this application.

Claims

1. A liquid transfer device, comprising: a liquid path mechanism, comprising a conduit assembly and a driving member for driving liquid to flow in the conduit assembly; a housing assembly, comprising a bottom plate and a housing with an accommodating space, wherein both the conduit assembly and the driving member are installed on the housing; a support frame, comprising a rotating self-locking structure and a telescopic structure; the telescopic structure is installed on the bottom plate through the rotating self-locking structure; the rotating self-locking structure is used for driving the telescopic structure to rotate relative to the bottom plate within a preset angle range, and locking the telescopic structure to stop rotating once it reaches a preset position; and an electrical assembly, comprising a control assembly installed in the accommodating space, and the control assembly is connected with the driving member.
2. The liquid transfer device of claim 1, wherein the liquid path mechanism is further installed at a bubble sensor on the housing, and the bubble sensor is used for detecting a liquid flow state in the conduit assembly.
3. The liquid transfer device of claim 1, wherein the electrical assembly further comprises a touch screen installed on the housing and connected with the control assembly; and/or the electrical assembly further comprises a switch button installed on the housing and connected with the control assembly.
4. The liquid transfer device of claim 1, wherein the electrical assembly further comprises a battery connected with the control assembly and a power supply interface connected with the battery and the control assembly.
5. The liquid transfer device of claim 1, wherein the housing comprises a main device housing installed on the bottom plate and having an opening, and a back plate installed at the opening position of the main device housing and connected with the bottom plate; the main device housing,

the back plate and the bottom plate enclose to form the accommodation space, and the control assembly is installed on the bottom plate and positioned in the accommodation space.

6. The liquid transfer device of claim 1, wherein the telescopic structure comprises: a support assembly, comprising a support plate with a sliding space, a slide rail arranged on an inner side wall of the sliding space, and a positioning hole arranged on the slide rail and communicated with the sliding space; a sliding assembly, comprising a sliding plate arranged in the sliding space and a positioning assembly arranged on the sliding plate, the sliding plate telescopically slides along the slide rail; and the positioning assembly is used for locking with the positioning hole when the sliding plate slides towards a first end of the slide rail, so that the sliding plate stops sliding, and the positioning assembly is also used for unlocking from the positioning hole when the sliding plate slides towards a second end of the slide rail.

7. The liquid transfer device of claim 6, wherein the positioning assembly comprises a resisting buckle arranged on the sliding plate, a mounting hole arranged on the sliding plate and communicated with the sliding space, and a first spring and a sliding block both installed in the mounting hole; an end of the sliding block abuts against the first spring, and another end of the sliding block extends into the sliding space from the mounting hole; a side of the sliding block near the first end of the slide rail is provided with a resisting surface; a side of the sliding block near the second end of the slide rail is provided with a guide surface; and the resisting buckle is provided with a through hole, and the sliding block passes through the through hole and extends into the sliding space from the mounting hole; the resisting surface of the sliding block is recessed, forming a clamping groove adapted to the resisting buckle.

8. The liquid transfer device of claim 6, wherein the rotating self-locking structure comprises: a rotating assembly, comprising a first rotating shaft and a second rotating shaft, the first rotating shaft and the second rotating shaft are respectively fixedly arranged on the opposite sides of the support plate and coaxially arranged; a bracket, the second rotating shaft is rotatably connected to the bracket; a lock assembly, comprising an elastic self-locking member slidably connected with the bracket and a torque hinge installed on the bracket, and the first rotating shaft is rotationally connected with the bracket via the torque hinge; the first rotating shaft is provided with a positioning groove, and the elastic self-locking member is provided with a locking column adapted to the positioning groove; the support plate rotates relative to the bottom plate around the first rotating shaft within a preset angle range; and the locking column is used for sliding into the positioning groove when the support plate rotates around the first rotating shaft to a preset position, so as to lock the support plate to limit the rotation of the support plate around the first rotating shaft, and the locking column is also used for unlocking the support plate when sliding out of the positioning groove.

9. The liquid transfer device of claim 8, wherein the elastic self-locking member comprises a lock buckle, a second spring and an unlocking pin; the locking column is arranged on the lock buckle; the bracket is provided with a first sliding hole and a second sliding hole parallel to each other, and the lock buckle is provided with a sliding column, the sliding column passes through and slides along the first sliding hole from a first side of the bracket; the unlocking pin passes through the second sliding hole from a second side of the bracket away from the first side and is connected with the lock buckle; and the second spring is sleeved on the unlocking pin and pressed between the bracket and the lock buckle.

10. The liquid transfer device of claim 8, wherein the first rotating shaft comprises a connecting segment connected with the support plate and a cooperating segment connected with the connecting segment; the cooperating segment is provided with an accommodating groove, the bracket is provided with an accommodating hole, an end of the torque hinge is interference-fitted in the accommodating hole, and another end of the torque hinge is interference-fitted in the accommodating groove; the cooperating segment is provided with an arc baffle, the arc baffle is inserted into the accommodating hole and slidably connected between the torque hinge and an

inner side wall of the accommodating hole; a stop block is arranged on the inner side wall of the accommodating hole; the stop block is used for limiting the first rotating shaft to drive the support plate to rotate when abutting against the arc baffle; and at least two positioning grooves are arranged on the cooperating segment; at least one positioning groove is aligned with the locking column when the stop block abuts against the arc baffle.
